3 A space shuttle of mass m orbits planet X of mass M and radius R, as shown in Fig. 3.1. The space station is at an altitude of $4R$ above the surface of the planet.

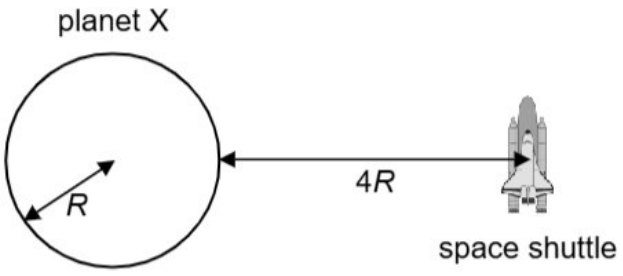


Fig. 3.1 (not drawn to scale)

(a) The linear velocity of the space shuttle is v.

(i) Explain why the space shuttle experiences a resultant force when it is in orbit.

.....

.....

..... [2]

(ii) Determine an expression for v^2 in terms of gravitational constant G, M and R.

v^2 = ... [2]

(iii) Hence, show that the kinetic energy E_K of the space shuttle in orbit is given by the expression:

$$E_K = \frac{GMm}{10R}$$

where G is the gravitational constant.

[1]

(b) Planet X has a mass of 7.8×10^{24} kg and radius of 6.8×10^{6} m. The space shuttle has a mass of 9.0×10^{4} kg.

Calculate the total energy of the space shuttle in its current orbit.

total energy = ... J [2]

(c) Later it is discovered that a nearby planet, known as planet Y, 6.0×10^{8} m away, affects the gravitational field of planet X, shown in Fig. 3.2.

Planet Y has the same radius but twice the mass of planet X.

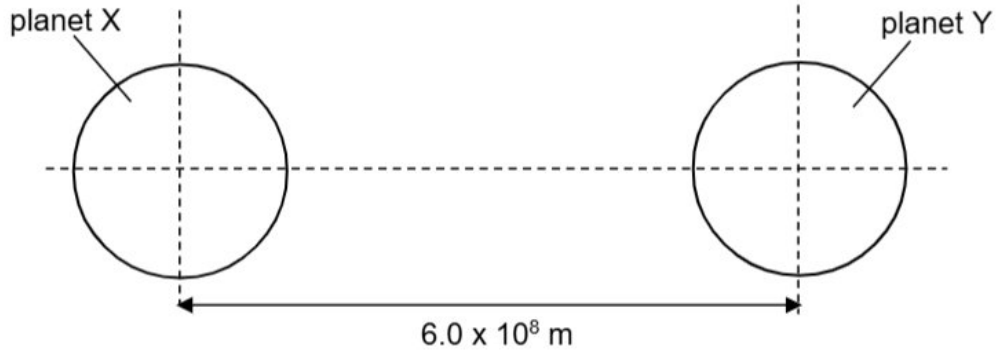


Fig. 3.2

(i) Point Z is a point along the line between the centres of the planets X and Y where the resultant gravitational field strength is zero.

Calculate the distance between Z and the centre of planet X.

distance = ... m [2]

(ii) The mission of the space shuttle in **(a)** is to reach the surface of planet Y.

Determine the total gravitational potential energy of the space shuttle if it reaches point Z.

total gravitational potential energy = ... J [2]

(iii) Hence, calculate the extra kinetic energy needed for the space shuttle to leave its current orbit of $5R$ around planet X and reach the surface of planet Y.

extra kinetic energy = ... J [2]

[Total:13]